How to take care of multicollinearity?

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- **Subset selection:** We can go for forward selection or backward deletion method to get a nested sequence of models and then select one of them using an appropriate Criterion (e.g., adjusted R^2, Mallows's C_p, AIC, BIC)

Ridge regression

Minimize $\|Y - X\beta\|^2$ subject to $\|\beta\|^2 \le C$ for some given $C > 0$.

This is equivalent to minimize

$$\|Y - X\beta\|^2 + \lambda(\|\beta\|^2 - C) \quad \text{where } \lambda > 0$$

or

$$\|Y - X\beta\|^2 + \lambda\|\beta\|^2 \quad \text{w.r.t. } \beta.$$

Let

$$\Psi(\beta) = (Y - X\beta)'(Y - X\beta) + \lambda\beta'\beta = Y'Y - \beta'X'Y - Y'X\beta + \beta'X'X\beta + \lambda\beta'\beta.$$

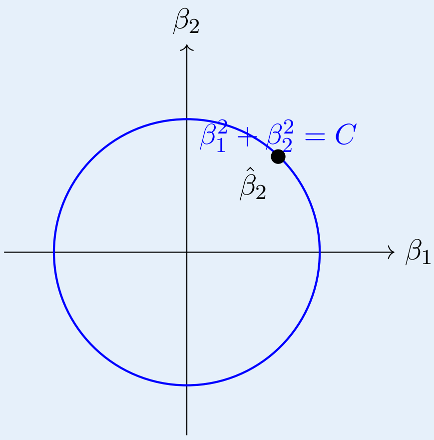
Then

$$\begin{aligned}
&\nabla\Psi(\beta) = -2X'Y + 2X'X\beta + 2\lambda\beta,\\
&\nabla\Psi(\beta) = 0 \implies X'X\beta + \lambda\beta = X'Y \implies (X'X + \lambda I)\beta = X'Y\\
&\qquad\quad \implies \beta = (X'X + \lambda I)^{-1}X'Y.
\end{aligned}$$

Notation:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{p1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \cdots & x_{pn} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}.$$

Geometric interpretation

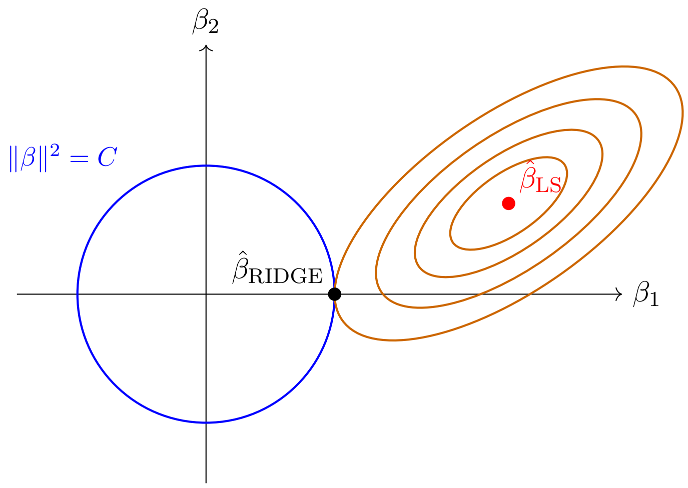


Result:

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{\text{LS}}\|^2 + (\beta - \hat{\beta}_{\text{LS}})'X'X(\beta - \hat{\beta}_{\text{LS}}).$$

The level curves of the residual sum of squares are ellipses centred at $\hat{\beta}_{\text{LS}}$. The ridge estimator is the point where the ellipse first touches the ball $\|\beta\|^2 = C$.

$$\|\hat{\beta}_{\text{RIDGE}}\|^2 = C, \qquad \hat{\beta}_{\text{RIDGE}} = (X'X + \lambda I)^{-1}X'Y.$$



For the two-variable illustration:

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i, \quad i = 1, 2, \ldots, n.$$

How to choose C (or λ) in practice?

Cross-validation method can be used for this purpose.

$$\min \|Y - X\beta\|^2 \quad \text{s.t.} \quad \beta'\beta \leq C.$$

$$\mathbb{E}(\hat{\beta}_{\text{ridge}}) = (X'X + \lambda I)^{-1}X'X\beta \neq \beta.$$

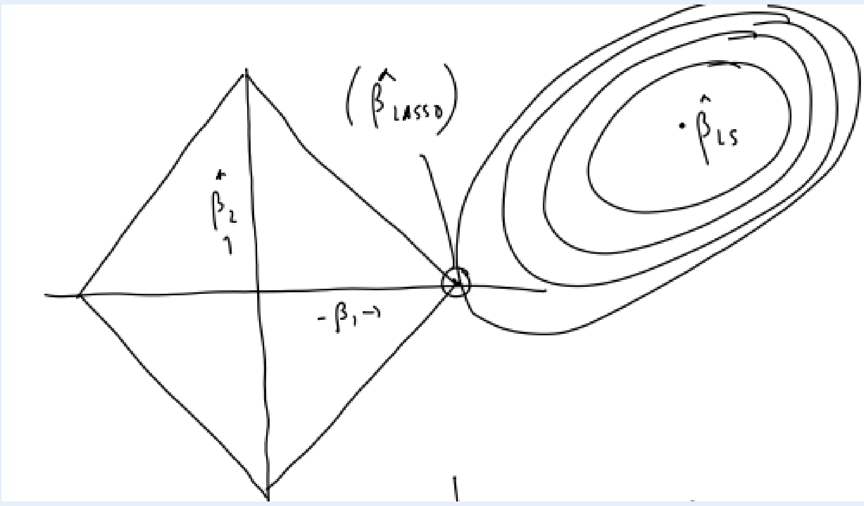
(The ridge estimator is biased.)

LASSO Regression

(Least Absolute Shrinkage and Selection Operator)

Here we minimize $\|Y - X\beta\|^2$ subject to $\|\beta\|_1 \le C$ for some $C > 0$.
(For a vector $z \in \mathbb{R}^p$, $\|z\|_1 = |z_1| + |z_2| + \cdots + |z_p|$.)

Geometry of the ℓ_1 constraint



$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{\mathrm{LS}}\|^2 + (\beta - \hat{\beta}_{\mathrm{LS}})' X' X (\beta - \hat{\beta}_{\mathrm{LS}}).$$

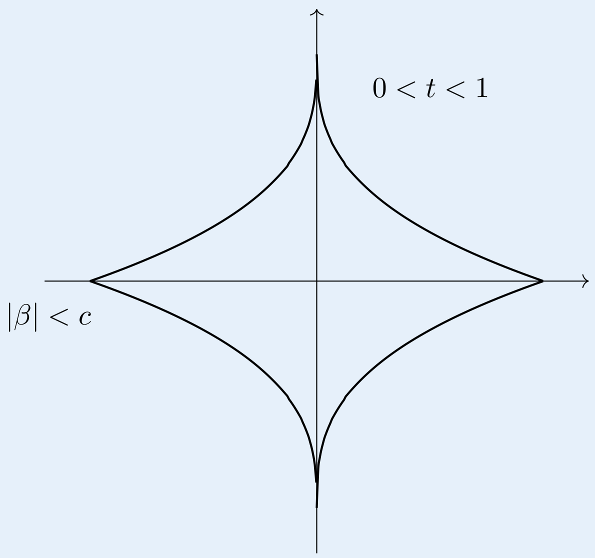
Minimization of

$$\|Y - X\beta\|^2 \quad \text{s.t.} \quad |\beta_1| + \cdots + |\beta_p| \le C$$

is equivalent to minimization of

$$\|Y - X\beta\|^2 + \lambda(|\beta_1| + \cdots + |\beta_p| - C).$$

This is a convex function.



For $0 < t < 1$,

$$|\beta_1|^t + |\beta_2|^t + \cdots + |\beta_p|^t \leq C$$

for some other t. However for $t < 1$, the objective function does not remain convex.

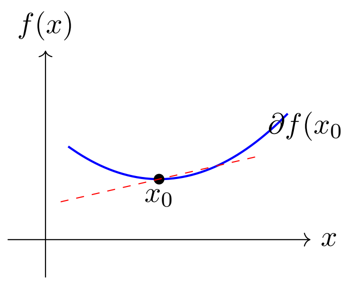
Sub-differential

The sub-differential of a function f at any point x_0 is defined as

$$\partial f(x_0) = \left\{c : f(x) - f(x_0) \geq c'(x - x_0)\ \forall\, x \in \mathbb{R}\right\}.$$

If f is convex and $f'(x_0)$ exists, then

$$\partial f(x_0) = \{m_0\} \quad \text{where} \quad f'(x_0) = m_0.$$



For $f(x) = |x|$,

$$\partial f(x_0) = \begin{cases} \{1\} & \text{if } x_0 > 0, \\ \{-1\} & \text{if } x_0 < 0, \\ [-1, 1] & \text{if } x_0 = 0. \end{cases}$$

If $0 \in \partial f(x_0)$, then x_0 is a minimizer of $f(\cdot)$.

$$\partial f(x_0) = \left\{c : f(x) - f(x_0) \geq c'(x - x_0)\right\}, \forall\, x \in \mathbb{R}^d.$$

$f : \mathbb{R}^d \longrightarrow \mathbb{R}$

Example

$f(x) = \|x\|$, the Euclidean norm

$$= \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}.$$

Show that

$$\partial f(x_0) = \begin{cases} \dfrac{x_0}{\|x_0\|} & \text{if } x_0 \neq 0, \\ k? & \text{if } x_0 = 0. \end{cases} \quad k = \{c : \|c\| \leq 1\}$$

(The set $\{c : \|c\| \leq 1\}$ is the unit ball in the dual norm.)

LASSO solution (for a special case where $X'X = I$)

Here we minimize

$$\Delta = \sum_{i=1}^{n} \Big(y_i - \sum_{j=1}^{p} \beta_j x_{ji} \Big)^2 + \lambda \big(|\beta_1| + |\beta_2| + \cdots + |\beta_p| \big).$$

If $\beta_k > 0$,

$$\frac{\partial \Delta}{\partial \beta_k} = -\sum_{i=1}^{n} 2 \Big(y_i - \sum_{j=1}^{p} \beta_j x_{ji} \Big) x_{ki} + \lambda.$$

Setting the derivative to zero:

$$\begin{aligned} \frac{\partial \Delta}{\partial \beta_k} = 0 &\implies -2 \sum_i y_i x_{ki} + 2\beta_k n + \lambda = 0 \\ &\implies -2\hat{\beta}_{k,\mathrm{LS}} + 2\beta_k + \lambda = 0 \end{aligned}$$

(using $X'X = I$, so $\sum_i x_{ji} x_{ki} = \delta_{jk}$ and the least-squares estimator simplifies accordingly).
More generally the sub-gradient condition is used when some coordinates are zero.

Sub-gradient definition (vector case)

$$\partial f(x_0) = \Big\{ c : f(x) - f(x_0) \geq c'(x - x_0) \ \forall\, x \in \mathbb{R}^d \Big\},$$

where $f : \mathbb{R}^d \to \mathbb{R}$.

$$\|Y - X\beta\|^2 = \sum_i \big(y_i - \beta_1 x_{1i} - \cdots - \beta_p x_{pi} \big)^2.$$

Since $X'X = I$,

$$\sum_i x_{ji} x_{ki} = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$